

Technical Research Report

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1. Document history

Table 1. Document History

Version	Date	Status	Author	Description	Remarks
0.1	03-09-2026	Draft Version	Casey Stijlaart	Draft Version	Initial document layout, non-camera sensing techniques chapter
0.2	04-09-2026	Draft Version	Casey Stijlaart	Draft Version	Finalisation of non-camera sensing techniques

2. Non-Camera Person Position & Height Sensing Techniques

2.1. Introduction

This chapter addresses the first technical research question defined in the project plan (see **Research Questions**): "Which combination of non-camera sensors provides sufficient 3D spatial resolution to distinguish adult and child height categories within a room-scale interactive space?"

Per the **Assignment**, the target system estimates a single person's real-time 3D position and height within the Holobox interaction space without cameras, exposing that estimate to the Holobox's Unreal Engine pipeline so the Virtual Human can adapt its behaviour accordingly. This chapter covers which non-camera sensing technologies exist, what they have actually been shown to do in the literature, and which module best fits the project's requirements and future extensibility: a literature-backed survey of candidate technologies, the criteria a viable approach must satisfy, a comparison against current hardware/cost data, and a recommended sensor choice with a staged de-risking plan. Embedded implementation detail, the on-device ML height-classification model, and the Unreal Engine output interface are addressed by other technical sub-questions; this chapter recommends what to use and how to validate that choice, not how the resulting embedded software is built. Per **Scope and Preconditions**, a fully polished prototype is out of scope, since the project is exploratory by nature; the recommendation here ends in a staged prototyping plan rather than a final hardware order.

Two aspects of the research question are narrowed for this stage: distinguishing adult and child height **categories** is treated as a possible later refinement rather than a hard requirement, since a general height estimate is sufficient for now (see Candidate Sensor Technologies); and multi-person tracking, a future-facing requirement in the Assignment, is addressed here only as an evaluation criterion (see Evaluation Criteria), not as a capability this stage builds. Both may be revisited later if time allows.

2.2. Research Approach

This chapter follows the Library and Lab strategies of the DOT-framework identified for this research question in **Research Methods**: available product analysis, literature study, document analysis, and bench experimentation.

2.2.1. Library: Literature Study

Candidate Sensor Technologies gathers and verifies peer-reviewed literature for each non-camera sensing family against RQ1's position/height requirement and, separately, against RQ4's multi-person active-speaker problem, in two separate passes: RQ1 sits in the radar/thermal/RF position-and-pose-estimation literature, while RQ4 sits closer to acoustic and vibrometry-based speaker localisation, so a single combined search would have missed sources specific to either.

2.2.2. Lab: Available Product Analysis

Alongside the literature, the same chapter narrows the literature-backed technology families down to specific, currently sourceable hardware modules, checked against live distributor listings (Digi-Key, Mouser, Adafruit, and others) rather than assumed pricing, since an unverified cost estimate is not a usable basis for a hardware decision or the budget-approval process in **Finance and Risks**. Literature backing and product availability are two different questions: a technique can be well-published without there being a specific, currently purchasable module that implements it at a reasonable price, or at all.

2.2.3. Lab: Bench Experimentation Strategy

The recommended path into bench experimentation commits to mmWave radar directly, managing implementation risk by splitting the build into capability-based phases rather than assuming the full pipeline works at once: position tracking first, then height estimation, then multi-person clustering, then speaker detection, each building on an already-working previous phase. This is deliberate: it isolates a failure to whichever capability is currently being added, rather than letting it surface somewhere in an untested, fully assembled system; the same mmWave hardware and UART/point-cloud/Unreal Engine bridge built in phase one is reused, not redone, in every phase after it; it follows the research-question sequence, validating RQ1 (position/height) before adding RQ4 (multi-person clustering, speaker detection); and it reflects feature priority, since RQ4's extensions improve the system without being required for the current single-person scope. A working fallback exists at each stage: if a later phase's capability underperforms, the earlier phase's version remains a functioning system rather than a dead end, mirroring the fallback the project's own risk register anticipates for both sensor selection and on-device ML performance.

2.3. Evaluation Criteria

Before diving into available sensor modules, it is of great importance to understand the requirements they need to meet. Thus the comparison can be set out against explicit, justified requirements rather than an implicit sense of what "good" looks like. Each criterion below is traced to either a technical sub-question in **Research Questions**, a key feature named in the **Assignment**, or a project constraint from **Finance and Risks**, rather than to a generic notion of sensor quality.

Table 2. Evaluation Criteria

Criterion	What it means here	Why it matters
Height-estimation capability	Must produce an actual height reading (alone or fused), not just (x, y) position.	RQ1's spatial-resolution requirement; a general estimate suffices for now, category classification is a later refinement.
Position accuracy and coverage	Position error and range must be usable at the Holobox's actual room-scale dimensions.	RQ1 covers (x, y) as well as height; a sensor that can't cover the space isn't usable regardless of height capability.
Real-time embedded feasibility	Interactive latency on an embedded platform: targeting <100 ms, hard ceiling ~1 s.	RQ2 requires low-latency embedded estimates; thresholds follow Nielsen's (1993) response-time limits, building on Miller (1968).
Multi-person extensibility	Shouldn't fundamentally break if a second person enters the space later.	RQ4 and the Assignment both name multi-person as a future-facing requirement.
Data minimisation / privacy	Output can't identify an individual; captures only what's needed for position and height.	Required by the ethical/legal sub-questions: visitors are children aged 8-12 (AVG/GDPR).
Literature and community backing	Prior published work shows the specific capability required, not just the technology in general.	Reduces the risk of committing project time to proving something never demonstrated at all.
Hardware cost	Cost of the sensor and any paired second sensor, checked against sourceable listings.	No project budget agreed yet with MindLabs, so cost is a real constraint.

NOTE

Sensor output rate is only one component of total pipeline latency, since processing, sensor fusion, and the Unreal Engine transmission step all add more

on top, but it sets a floor worth checking early.

These criteria are not weighted equally, and are not applied as a hard pass/fail filter. Height-estimation capability and data minimisation are treated as closer to hard requirements, since they follow directly from RQ1 and the ethical/legal sub-questions; the remaining criteria (sensor combination, cost, power/footprint) are weighed against each other in the chapter that follows, since a technology that is merely more expensive or harder to pair with a second sensor is not disqualified the way one with no height precedent or an identifying output effectively is.

2.4. Candidate Sensor Technologies

This chapter surveys non-camera sensing technologies capable of device-free person detection, localisation, and (where the literature supports it) height estimation, evaluated against the technical sub-questions in **Project Plan - Research Questions**: primarily sufficient 3D spatial resolution for height, embedded-platform feasibility, and the data-minimisation/privacy constraints that follow from the primary visitors being children aged 8-12. A general height estimate is sufficient for now; distinguishing adult and child height categories is a possible later refinement, not a hard requirement.

Each technology is covered in two parts: what the literature has actually shown it can do, and, for the technically viable candidates, how that holds up against the criteria set out earlier, specific hardware, cost, and pros/cons, closing in a comparison summary, criteria matrix, and recommendation. Ngamakeur et al. (2020) frame this same comparison at a broader scale, structuring device-free sensing into a pipeline (**Figure 1**) used below to organise what each technology has actually been shown to do, rather than what it could in principle do.

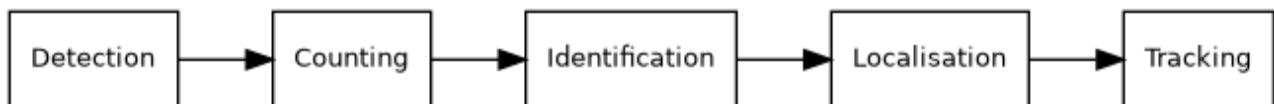


Figure 1. Device-free Pipeline

2.4.1. Individual Position & Height Localisation

This section covers single-person, device-free position and height sensing, the core of RQ1.

Time-of-Flight & LiDAR

2D LiDAR measures range by timing a laser pulse's reflection while scanning a single horizontal plane, producing an (x, y) point cloud at the unit's mounted height; well suited to in-plane tracking, but a single fixed-height plane does not by itself yield a height measurement (would need multiple units at different heights, a tilted scan, or a full 3D ToF/LiDAR unit).

Within its native (x, y) role, 2D LiDAR performs well: Bouazizi et al. (2024) fuse multiple units with a ConvLSTM model and show tolerance to furniture occlusion, and Enoki et al. (2024) show waist-height contour data alone is enough for person identification and activity estimation via DBSCAN clustering, indicating the modality carries enough per-person shape information even from one plane. For RQ4, Alam et al. (2021) document "crossover ambiguity" as 2D LiDAR's recurring multi-person failure mode when tracks pass close together, the concrete risk to plan around later. The representative module, the RPLIDAR A2M8, scans at 5-15 Hz (typically 10 Hz) per Slamtec's datasheet, the limit on full (x, y) frame availability regardless of its faster ~4000 samples/sec per-sample rate.

Implication for RQ1: 2D LiDAR alone does not produce a height estimate; it would need pairing with a second height-sensing modality, carried forward as **2D LiDAR + Thermal Fusion** below.

Pros:

- Strong in-plane position accuracy and furniture-occlusion tolerance when multiple units are fused (Bouazizi et al.).
- A single scan plane already carries enough per-person shape information for identification/activity classification (Enoki et al.).
- Mature, widely available consumer/robotics hardware (RPLIDAR family) with broad community support.

Cons:

- No height reading from a single scan plane; needs a second sensor or added mechanical complexity to get one.
- "Crossover ambiguity" is a documented multi-person failure mode when tracks pass close together (Alam et al.).
- 5-15 Hz scan rate is on the slower end of the candidates reviewed.

mmWave Radar

mmWave radar has an extensive body of research literature covering both position tracking and skeletal/joint-level pose estimation, and its point-cloud output is inherently non-identifying (no recognisable imagery), aligning with the sub-questions on minimising personally identifiable data from child visitors.

For position tracking, Huang et al. (2021) combine clutter removal, clustering, and Kalman filtering for real-time multi-person tracking on a Raspberry Pi 4, and Pegoraro and Rossi (2022) report position error down to roughly 7.6 cm. More directly relevant to height estimation is the pose-estimation literature: Zhu et al.'s ProbRadarM3F (2024) recovers 14 body keypoints from radar point clouds, and Zheng et al. (2025) survey related keypoint work; since keypoints include head/shoulder joints, this is the most direct existing path to an actual height estimate rather than a size proxy, backed by camera-free radar datasets with joint ground truth (Su et al. (2024), Rahman et al. (2024)). On latency, reported tracking papers give broadly similar tens-of-milliseconds frame periods on different radar chips (15-20 fps), and the IWR6843 chip itself supports frame rates up to ~76 Hz per TI's datasheet, though typical demo configurations run at 10-20 Hz; see Evaluation Criteria for the thresholds this is checked against.

Implication for RQ1: mmWave radar's literature includes published keypoint/pose-level output alongside demonstrated real-time embedded operation, and its non-identifying output profile fits the privacy sub-questions well.

Available product analysis narrows this to a 60 GHz point-cloud radar module built around the TI IWR6843 chip, over 24 GHz presence sensors (binary presence flag only, no point cloud) and undocumented generic modules lacking reference software. Several IWR6843 evaluation-board variants exist at different price points and field-of-view characteristics; **Module Selection** compares them and picks one to carry forward.

Pros:

- Strongest literature base of any candidate: real-time multi-person tracking and keypoint-level pose/height estimation.
- Demonstrated real-time operation on a Raspberry Pi 4.
- Point-cloud output is inherently non-identifying, fitting the project's data-minimisation requirements.

Cons:

- By far the most expensive candidate (€160-185 for the IWR6843AOPEVM variant compared in **Module Selection**).
- Height estimation still requires extra work: a blob-size heuristic or a keypoint-extraction model, not a single established method.
- No project budget has yet been agreed with MindLabs, so this cost is a real, unresolved risk.

Wi-Fi CSI Sensing

Wi-Fi Channel State Information (CSI) sensing infers presence and movement from fluctuations in the wireless channel caused by a body moving through it, using only commodity Wi-Fi hardware, but the gathered literature is consistent that it trades away spatial resolution and stability: CSI is sensitive to environmental changes (furniture, multipath, other RF sources).

IndoTrack (2017) is the foundational device-free CSI tracking paper in this space; more recent work narrows toward multi-person settings, e.g. Jiang and Yuan's GMM/weighted k-NN fingerprinting approach (2025), Shen et al.'s time-selective RNN for busy-room detection (2023), and Chung et al.'s WiRiS (2023), which jointly solves people counting and localisation via a transformer over RIS-assisted CSI. Reported update rates vary by an order of magnitude across otherwise comparable commodity setups, from 10 Hz (Shen et al., capped by their router) to a reported 100 Hz (IndoTrack, unconfirmed against the primary paper), an additional latency risk on top of the lack of height output.

Implication for RQ1: none of the gathered sources report height or pose-level output, and the modality's environment-sensitive resolution limits its usefulness for height estimation on its own; it remains worth carrying forward as a low-cost presence/occupancy option.

Pros:

- Effectively free: reuses existing Wi-Fi infrastructure.
- Multi-person and multi-room literature exists (Time-Selective RNN, WiRiS).
- Device-free output with no recognisable imagery.

Cons:

- No published height-estimation precedent at all, the single biggest gap against RQ1.
- More sensitive to environmental changes than radar or thermal.
- No demonstrated real-time embedded deployment in the sources reviewed.

Ultrasonic Ranging

Active ultrasonic ranging measures distance via time-difference-of-arrival (TDOA) or time-of-flight on emitted pulses, sidestepping the "room full of talking people" interference problem a passive-listening approach would face.

Caicedo and Pandharipande (2012) use a beamforming array with moving-target-indicator processing and report it outperforming PIR sensors for occupancy detection; Segers et al. (2015) track up to four transmitters at 1-2cm accuracy on an FPGA using frequency-shift-keyed ultrasound, and Holm and Nilsen (2010) demonstrate TDOA-based positioning from a transmitter array. Neither Segers et al. nor Caicedo and Pandharipande report an update rate or latency figure, so latency here is an open question rather than a documented one.

Implication for RQ1: ultrasonic ranging gives decent range accuracy, but azimuth resolution is limited, full coverage needs multiple synchronised nodes (reintroducing the multi-sensor complexity active signals were meant to avoid), and no gathered source addresses height or elevation at all; a vertically-oriented array would be needed for a z-axis reading, which is not demonstrated in the evidence base.

Pros:

- Sidesteps the "room full of talking people" interference problem passive listening would face.
- Demonstrated real-time operation on embedded hardware (an FPGA, per Segers et al.).
- Reasonable literature base for ranging and positioning.

Cons:

- No gathered source addresses height or elevation.
- Full coverage needs multiple synchronised nodes, reintroducing multi-sensor complexity.
- One of the pricier custom-built options (€200-400).

Infrared / Thermal

Low-resolution thermal/thermopile arrays measure a grid of per-pixel temperature readings; a person would appear as a warm blob against the cooler background, and the same low pixel count that limits resolution and range is itself a privacy advantage, since it's far below what's needed to identify a person or resolve recognisable imagery.

Most directly relevant to RQ1, Naser et al. (2021) explicitly extend thermal-array distance estimation to height measurement using a neural network, and should be read first. Feasibility on embedded hardware is demonstrated by Shubha and Shastrimath (2022), who report 99% occupancy accuracy on a Grid-EYE/AMG8833 array on a Raspberry Pi; Shetty et al.'s foundational Grid-EYE work (2017) is extended by Klir et al. (2025) for multi-person localisation (relevant to RQ4), and Trofimova et al. (2017) address detection robustness via adaptive background estimation. Naser et al.'s sensor supports a 0.5-64 Hz sampling rate; the specific Grid-EYE/AMG8833 breakout identified for procurement is coarser, exposing only a 10 Hz or 1 Hz frame-rate register.

Implication for RQ1: thermal array has a source explicitly performing height estimation from sensor data, and its low resolution doubles as a privacy strength rather than only a limitation; the main tradeoff to carry forward is its effective range, checked against the interactive space's actual dimensions.

Sensor hardware cost is substantially lower than mmWave, in the order of 2 x €40-55 excluding the embedded compute platform, confirmed at €49.60 (incl. VAT) via [Kiwi Electronics](#) (NL), and the I2C interface is simpler to bring up than a radar UART/configuration pipeline. That price is for the AMG8833's narrow ~60° field of view, sufficient only for a small room; wider coverage without narrowing the FoV would normally mean mounting overhead, which is not an option for the Holobox's fixed installation, so room-scale coverage instead means the wider-FoV MLX90640 (110° FoV) variant, €75.01 (incl. VAT) at [Kiwi Electronics](#) (NL). Wherever thermal is paired with a room-scale primary sensor in [Combined Sensor Approaches](#) below, two units are needed for full room coverage rather than one; since coverage is then split between two sensors, neither one needs the wider (pricier) FoV on its own, so two of the cheaper narrow-FoV AMG8833 units (roughly €100 total) are the cost basis used there, not the single wide-FoV part.

Pros:

- The only technology reviewed with a source performing height estimation directly from sensor data (Naser et al.).
- Cheapest viable sensor-only option found (€40-55 narrow-FoV; €70-75 for the wide-FoV variant needed for room-scale coverage).
- Demonstrated real-time embedded operation on a Raspberry Pi.
- Low pixel count doubles as a privacy strength: output cannot resolve recognisable imagery.

Cons:

- Shorter effective range than mmWave, constraining it to smaller rooms unless the pricier wide-FoV variant is used.

- Lower spatial resolution overall, which needs checking against the Holobox's actual dimensions.

Capacitive & Floor-based Sensing

Capacitive / Electric-Field Sensing

Two device-free variants exist: proximity-plate sensing (measuring distance to a body via a change in capacitance) and floor-embedded capacitive electrodes (requiring a person to stand on the sensed area). Ramezani Akhmareh et al. (2016) and Bian et al.'s survey (2024) cover the design space broadly; Faulkner et al.'s CapLoc (2020) is a floor-embedded mat evaluated in the EvAAL indoor-localisation competition, and Fu et al.'s passive Linoc system (2017) is reported as less noise-prone than active designs. Range is the main constraint: plate sensors top out around 0.3-1m, so whole-room coverage needs a floor grid or many nodes, and the signal is inherently presence/proximity rather than spatial, so it is not naturally suited to a height reading at all.

Implication for RQ1: no reported height output, and short range makes whole-room coverage costly without a dense sensor grid.

Floor-based (Pressure) Sensing

Floor-based sensing (pressure, piezoresistive, or capacitive-floor mats) is inherently a 2D floor-plane measurement, but because each footstep is a spatially discrete event, it naturally disambiguates multiple people. Gatto et al. (2015) use a 64x64 pressure-switch array per tile; Al-Naimi et al. (2014) fuse pressure-floor tracking with PIR sensors; Feng et al. (2015) use a fiber-optic floor grid for multi-target localisation, and Ibara et al. (2013) show gait-based individual identification from stride length and direction.

Implication for RQ1: floor sensing trivially solves "how many people, where," since footsteps are naturally separated events, but its weaknesses are installation cost and a complete absence of height information; carrying it forward would mean pairing it with a separate height-sensing modality.

Pros:

- Floor-based variants trivially solve "how many people, where."
- Solid literature base for both sub-variants.
- Inherently non-identifying (presence/proximity or pressure, not imagery).

Cons:

- No height signal at all from either sub-variant.
- Capacitive sensing has short range, needing a dense grid for whole-room coverage.
- Floor-based sensing requires physical installation under the existing floor; cost can run into the thousands of euros.

Ultra-Wideband (UWB)

Device-free UWB sensing (IR-UWB) uses very short radio pulses across a wide bandwidth to estimate range and, via signal processing or ML, detect and count people without a carried tag. Evidence is limited to a single source: Njanda et al. (2024) apply ML to IR-UWB radar for people counting, reporting 99% accuracy with an ANN classifier; the representative module (Novelda Xethru X4) has no single fixed frame rate, trading directly against SNR/processing gain, and Njanda et al.'s paper does not confirm which hardware or frame rate they used.

Implication for RQ1: the evidence base is currently too thin to draw a conclusion, as no gathered source addresses height or pose-level output; flagged for a further literature pass rather than ruled out on one source.

Pros:

- People-counting literature exists (99% accuracy, ANN classifier).
- Radar-based output is inherently non-identifying.

Cons:

- No height or pose-level literature at all; flagged as unproven.
- Only a single source exists, too thin an evidence base to draw a conclusion from.
- The representative module is no longer available through an authorised distributor, making cost unverifiable.

2.4.2. Multi-Person: Identifying the Active Speaker

Everything above addresses single-person position and height. A separate problem, relevant to RQ4's extensibility to multi-person scenarios, is that once several people are present in the interactive space, the system also needs to determine which one is currently speaking to the Virtual Human, without a camera, and without relying on people carrying a microphone or tag. None of the sources below have been tested in this project's specific context (children's voices, a room-scale interactive installation), so every option here is flagged for a dedicated literature and bench-testing pass rather than treated as settled.

mmWave Radar: Resolution Limits for Counting, Height, and Speaker Separation

Whether the IWR6843 reports two people as distinct targets, rather than merging them into one detection, depends on the radar's angular and range resolution, not on a single blanket "multi-person" capability. Two people need roughly 1.7-2.9m of horizontal separation, or a height difference of ~1.7m (e.g. one seated, one standing), before angular resolution reliably reports them as separate; range separation (front-back) is far finer, ~4-5cm, and follows automatically from time-of-flight rather than needing angular discrimination. Speaker identification does not share this constraint: it relies on Doppler micro-motion detection of mouth/chest movement during speech (the same mechanism as the mmWave vibrometry work below), so the radar can identify who is speaking regardless of how close together people are standing. For a room roughly 5m wide with 2-3 people scattered across it, ordinary spacing satisfies all three requirements together (counting, height, speaker ID); the failure case to plan around is 4+ people clustered tightly, where horizontal separation drops below the 1.7-2.9m threshold and tracks become ambiguous, the same failure mode as 2D LiDAR's "crossover ambiguity" above.

mmWave Vibrometry

mmWave radar's vibrometry capability extends to sound itself: Ozturk et al. (2021) show mmWave radar can reconstruct audio from a sound source's surface vibrations, including a human throat, even through soundproofing, and Xu et al.'s WaveEar (2019) uses a low-cost mmWave probe to first localise a speaker among multiple people, then steer toward their throat, solving "who is speaking" and "what are they saying" from the same sensor.

Implication for RQ4: mmWave vibrometry can localise a speaker among multiple people and recover their voice from the same radar signal.

Wi-Fi CSI Sensing

Wi-Fi CSI reflections from mouth/lip movement during speech are detectable using only commodity Wi-Fi: Wang et al.'s WiHear (2014) uses directional antennas to sense multiple people's talk simultaneously via MIMO, and Du et al.'s WiTalk (2018) reports over 92% accuracy discerning syllables from CSI dynamics alone.

Implication for RQ4: Wi-Fi CSI can detect and analyse mouth-movement reflections associated with speech using only commodity hardware.

UWB Radar

IR-UWB radar can remotely detect lip/jaw motion associated with speech: Shin and Seo (2016) extract these features from IR-UWB radar signals for contactless silent speech recognition.

Implication for RQ4: IR-UWB radar can detect articulator motion associated with speech from radar reflections alone.

Laser Doppler Vibrometry

A laser Doppler vibrometer (LDV) measures skin-surface vibration at a distance via Doppler

shift, picking up throat vibration during speech even in noisy conditions; Avargel and Cohen (2011) combine an LDV sensor aimed at the larynx with a standard microphone to enhance speech quality under transient noise.

Implication for RQ4: LDV has high immunity to ambient noise, but requires the laser aimed precisely at the throat, and several published setups rely on a camera to aim and track the target, which would need a non-camera aiming mechanism here.

Microphone Array (Passive Acoustic)

Berghi and Jackson (2023) use a microphone array with a convolutional recurrent neural network to localise an active talker's position from multichannel audio alone, requiring no camera or radar hardware at all.

Implication for RQ4: a microphone array can localise an active talker's position from audio alone, with no line-of-sight or radar hardware requirement.

2.4.3. Combined Sensor Approaches

The individual technologies above are also weighed as sensor combinations, where pairing two of them addresses a gap neither has alone.

mmWave + Thermal Fusion for Height

The project's own risk analysis anticipates that a single sensor may not reliably discriminate height across the full range, and names combining "two or more complementary sensors (fusion)" as the planned fallback. The concrete instantiation of that plan is to add the infrared/thermal array as a second, height-specific sensor alongside mmWave, rather than replacing mmWave outright: mmWave continues to provide position and person tracking, while the thermal array's independent height reading, backed by Naser et al. (2021), is used to confirm, or substitute for, mmWave's own height estimate.

This is a narrower, lower-risk form of fusion than **2D LiDAR + Thermal Fusion** below: that combination requires fusing two independent position-sensing streams from scratch, whereas this option keeps mmWave as the sole position sensor and only adds a second signal for the specific sub-problem, height. The main added implementation work is registering the thermal array's field of view against the mmWave-detected person's position, rather than fusing two full tracking pipelines. A single thermal array does not cover the Holobox's whole room from one wall/corner mounting position (ceiling-mounting to compensate is not an option here); two units, positioned to jointly cover the room, are needed instead. Since coverage is then split between the two, neither needs the wider (pricier) FoV on its own: two AMG8833 units at roughly €50 each, €100 total, cover the room more cheaply than one wide-FoV part would alone. Combined sensor-only hardware cost is therefore the mmWave entry in **Section 2.4.4** plus two thermal units, roughly €260-285, plus a single shared compute platform.

Pros:

- Combines mmWave's strong position/multi-person tracking with thermal's direct height-estimation precedent.
- Gives a concrete fallback if mmWave's own height estimate underperforms, without discarding the mmWave investment.
- Reuses the same thermal hardware already evaluated as a standalone candidate above, rather than adding a third, unfamiliar sensor.

Cons:

- Requires registering each thermal unit's field of view against the mmWave-detected person's position: real added implementation work, doubled across two units.
- Needs two thermal units for full room coverage, not one, pushing combined cost to €260-285 total, on top of the pricier mmWave component.
- No single published source tests this exact combination; each half is validated separately, not together.

2D LiDAR + Thermal Fusion

As with the other thermal-fusion options above, the thermal side here is two AMG8833 units (~€100 total), splitting room coverage between them rather than relying on one pricier wide-FoV MLX90640 or a single narrow-FoV unit that falls short on its own. The representative LiDAR module itself, the RPLIDAR A2M8, also now shows as discontinued/out of stock at major distributors (e.g. Seeed Studio), an added sourcing risk on top of the concerns below.

Pros:

- 2D LiDAR contributes strong in-plane position accuracy and occlusion tolerance (Bouazizi et al.).
- Thermal contributes the same direct height-estimation precedent used elsewhere in this chapter (Naser et al.).

Cons:

- The most expensive combination reviewed in this chapter (€350-390, sensor-only), driven up further by needing two thermal units.
- Combining two independent position-sensing streams from scratch introduces synchronisation and fusion complexity flagged as a disproportionate implementation risk for this project's timeline, per **Timeline**.
- The representative RPLIDAR A2M8 module is discontinued at major distributors, an unresolved sourcing risk specific to this combination.
- Unlike the mmWave + Thermal option, this does not reuse an already-working single-sensor pipeline.

2.4.4. Comparison Summary

Table 3. Candidate Sensor Comparison Summary

Technology	Representative Module	Approx. Hardware Cost (sensor only)	Height Data in Literature	Documented Latency / Update Rate
mmWave Radar	TI IWR6843AOPEVM (see Module Selection for the variant comparison)	€160-185	Yes, keypoint-based	IWR6843 datasheet: up to ~76 Hz, typically 10-20 Hz in practice. Cited tracking papers report 15-20 fps on different radar chips.
Infrared/Thermal Array	AMG8833 breakout, Kiwi Electronics (NL)	€40-55	Yes, direct	AMG8833 datasheet: 10 Hz or 1 Hz selectable.
2D LiDAR + Thermal (sensor fusion)	RPLIDAR A2M8 (discontinued at major distributors) + 2x AMG8833, Kiwi Electronics (NL)	€350-390	Partial, via fusion	RPLIDAR A2M8 datasheet: 5-15 Hz scan rate, typically 10 Hz.
Wi-Fi CSI Sensing	None. It reuses existing commodity Wi-Fi hardware	Approx. €0 (existing infrastructure)	None found	Hardware-dependent: 10 Hz (Shen et al., 2023) up to a reported 100 Hz (IndoTrack, unverified against primary source).
Ultrasonic Ranging	Custom transducer array (e.g. SPM0404UD5 MEMS elements, per Segers et al. (2015))	€200-400 (rough estimate for custom-built hardware, not a single purchasable SKU)	None found	Not reported in gathered sources.

Technology	Representative Module	Approx. Hardware Cost (sensor only)	Height Data in Literature	Documented Latency / Update Rate
Capacitive & Floor-based Sensing	Custom-built capacitive/pressure floor mat. There is no off-the-shelf module	€80-2000+, depending on variant (rough estimate for custom-built hardware)	None found	Not reported in gathered sources.
Ultra-Wideband (UWB)	Novelda Xethru X4 (per Njanda et al. (2024))	Unverifiable: no longer available through an authorised distributor; only found via secondary-market listings, so not treated as a sourceable cost	Unproven	Not a fixed spec: trades off against SNR, up to a 40.5 MHz pulse repetition frequency ceiling per the Xethru X4 datasheet.

Figures above are sensor/module hardware only. All candidates other than Wi-Fi CSI also need an embedded compute platform to run the sensing pipeline; that cost is excluded here since it is a shared requirement across candidates rather than something that distinguishes one sensor choice from another. Latency figures are the sensor's own documented output rate, not a measured end-to-end pipeline latency; see the caveat already noted in Evaluation Criteria.

Table 4. Criteria Comparison Matrix

Technology	Height Est.	Embedded Real-Time	Multi-Person	Fusion-Ready	Privacy	Literature	Price Range (sensor only)
mmWave Radar	✓	✓	✓	✓	✓	✓	€160-250
Infrared/Thermal Array	✓	✓	✓	✓	✓	✓	€80-110
mmWave + Thermal Fusion	✓	✓	✓	✓	✓	□	€200-300
2D LiDAR + Thermal Fusion	✓	□	✓	✓	✓	✓	€350-390
Wi-Fi CSI Sensing	□	□	✓	□	✓	✓	~€0 (existing infrastructure)
Ultrasonic Ranging	□	✓	□	□	✓	✓	€200-400
Capacitive & Floor-based Sensing	□	□	✓	□	✓	✓	€80-2000+
Ultra-Wideband (UWB)	□	□	✓	□	✓	□	Unverifiable

2.4.5. Recommended Approach: mmWave, Phased by Capability

On the criteria set out earlier, mmWave radar comes out ahead: it is the only approach with keypoint-level pose literature, it extends to the RQ4 speaker-identification problem via the same sensor, its output is inherently non-identifying, and it has the strongest literature base of any candidate reviewed. The infrared/thermal array remains the only approach with a direct height-estimation precedent and by far the cheapest to acquire, but on technical merit mmWave is the stronger choice, and it is recommended directly rather than deferred behind a cheaper sensor. This is a recommendation of the mmWave modality; which specific IWR6843 evaluation-board variant to build with is a separate, narrower decision, addressed next in [Module Selection](#).

mmWave is, by a wide margin, still the most expensive candidate reviewed (roughly three to four times the thermal array's sensor cost), and no project budget has yet been agreed with MindLabs. That cost is treated as a precondition to clear with the company mentor before hardware is ordered, not as a reason to build something else first. What is staged instead is the implementation itself: rather than assuming position tracking, height estimation, multi-person clustering, and speaker detection all work together on the first attempt, the mmWave pipeline is built up one capability at a time, each on top of an already-working previous phase. This still isolates risk (a problem in one capability does not require redoing the whole pipeline) without introducing a second sensor and a mid-project hardware swap. If mmWave's own height estimate underperforms once measured, [mmWave + Thermal Fusion for Height](#) remains the documented fallback.

The concrete phasing of this capability build-up is carried into the Conclusion.

2.5. Module Selection

Candidate Sensor Technologies recommends mmWave radar as the sensing modality, built around the TI IWR6843 chip. That chip ships on several evaluation-board variants at different price points and field-of-view characteristics, given below as azimuth (angular coverage side-to-side) / elevation (angular coverage top-to-bottom, i.e. floor-to-head reach), all sharing the same TI mmWave SDK/point-cloud software stack and all operable standalone over USB (none strictly require the separate MMWAVEICBOOST carrier card for point-cloud demos). This chapter compares those variants directly and picks one. None of these professional evaluation boards are stocked by Dutch hobbyist-electronics shops (Kiwi Electronics, Antratek, TinyTronics); the actual sourcing path for a Dutch buyer is DigiKey's or Mouser's Netherlands storefronts, both of which ship locally and quote in EUR, which is what the links and prices below reflect.

- **IWR6843LEVM**: TI's cost-optimised FR4-antenna variant, €140-160 at current **Digi-Key NL** listings, with a 120° azimuth / 80° elevation field of view. The cheapest usable (i.e. not bare-chip) option found.
- **IWR6843ISK**: the standard long-range antenna plug-in module, €175-265 at current **Mouser NL** listings, with a narrower 108° azimuth / 44° elevation field of view tuned for range over coverage. Better documented in some community projects, but its narrow elevation beam works against a room-scale height-estimation use case rather than for it.
- **IWR6843ISK-ODS**: a wide-field-of-view (120° azimuth / 120° elevation) antenna variant, etched on a Rogers substrate, that TI markets specifically for overhead/ceiling-mounted people counting and tracking, €195-215, ~12 m range. Better suited than the plain ISK if the Holobox sensor ends up ceiling-mounted, since that use case is exactly what this variant's antenna is designed for.
- **IWR6843AOPEVM**: an antenna-on-package (AoP) variant, with the antenna integrated directly into the chip package rather than etched onto a separate PCB, €160-185 at current **Mouser NL** / **Digi-Key NL** listings. Matches the ISK-ODS's wide 120° azimuth / 120° elevation field of view in a smaller (~65×60mm), standard (non-overhead-specific) form factor, at roughly the same price as the plain LEVM.
- **Infineon BGT60TR13C** eval board: a comparable, EU-sourced 60 GHz alternative outside the TI ecosystem, €230.

Table 5. mmWave EVM Variant Comparison

Module	Antenna	Field of View (Az/EI)	Approx. Price	Best fit
IWR6843LEVM	Cost-optimised FR4 PCB antenna	120°/80°	€140-160	Cheapest usable option; adequate but not class-leading elevation coverage.
IWR6843ISK	Long-range PCB antenna	108°/44°	€175-265	Narrow elevation FoV suits a defined industrial zone, not room-scale height coverage.
IWR6843ISK-ODS	Wide-FoV PCB antenna (Rogers substrate)	120°/120°	€195-215	Purpose-built for ceiling/overhead mounting specifically.
IWR6843AOPEVM	Antenna-on-package (integrated into chip)	120°/120°	€160-185	Same wide coverage as ISK-ODS, more compact, mount-agnostic, and no external antenna to design around.
Infineon BGT60TR13C	PCB antenna (EU-sourced)	Not independently verified	€230	Non-TI fallback if TI sourcing becomes a problem.

On field of view, the ISK's narrow 44° elevation beam is a poor match for this project: it is tuned for a defined industrial zone at range, not for seeing a person from roughly floor to head height across a room, which is what the height-estimation use case actually needs. The ISK-ODS and AOPEVM both offer the wide 120° elevation FoV that fits room-scale height coverage, at similar prices (€160-185 to €195-215) and near the LEVM's own price point.

2.5.1. Recommendation: IWR6843AOPEVM

Between the ISK-ODS and the AOPEVM, this chapter recommends the **IWR6843AOPEVM**: it matches the ISK-ODS's field of view without being purpose-tuned to overhead-only mounting (leaving mounting position flexible while the Holobox's sensor placement is still being finalised), its antenna-on-package design removes the external antenna as a separate design/tuning problem entirely (a real simplification for a small embedded team, and a lower-risk path if the module is ever redesigned onto a custom production board), and its more compact ~65×60mm footprint eases fitting inside the Holobox's existing enclosure, per the power-consumption-and-footprint criterion in Evaluation Criteria. The trade-off is that a 120° elevation beam picks up more off-axis/peripheral reflections than the ISK's narrower beam, which the clutter-removal and clustering step (following Huang et al.) already in the pipeline is expected to absorb, so it is not treated as disqualifying.

This module choice does not change the mmWave modality recommendation or its cost bracket (€160-185 sensor-only, in the same range as the plain LEVM), so the phased-build plan and fallback options set out in Candidate Sensor Technologies' Recommended Approach carry forward unchanged.

2.6. Conclusion

Candidate Sensor Technologies weighed the technically viable sensors and sensor combinations against the criteria in Evaluation Criteria. On that basis, mmWave radar is recommended directly: it is the only approach with keypoint-level pose literature for height estimation, it extends to the RQ4 active-speaker problem via the same sensor, its output is inherently non-identifying, and it has the strongest literature base of any candidate reviewed. Within the TI IWR6843 family, Module Selection picks the **IWR6843AOPEVM**: its antenna-on-package design gives the wide 120° elevation field of view needed for room-scale height coverage without a separate antenna to design around, in a compact, mount-agnostic form factor at roughly the same price as the cheaper plain LEVM variant.

Rather than deferring mmWave behind a cheaper sensor, implementation risk is managed by splitting the mmWave build itself into capability-based phases: position tracking first, then height estimation, then multi-person clustering, then speaker detection, each building on an already-working previous phase. This still isolates risk (a problem in one capability does not require redoing the whole pipeline) without introducing a second sensor and a mid-project hardware swap. mmWave remains, by a wide margin, the most expensive candidate reviewed, and no project budget has yet been agreed with MindLabs, so mentor approval of the full mmWave hardware cost is a precondition for this plan, not a detail to resolve later.

Sensor fusion, combining mmWave (or 2D LiDAR) with a second thermal sensor, was deliberately not adopted as the starting recommendation, even though the project's own risk analysis anticipated it as a likely need. mmWave alone already has a plausible, literature-backed path to a height estimate via keypoint/pose extraction, so a second sensor is not needed to attempt a first working pipeline; fusion is kept as a documented fallback for if that estimate underperforms once measured, not a precondition for starting. Committing to fusion upfront would also mean paying its cost and complexity before that need is established: none of the three fusion combinations evaluated has been tested together in the literature (each half is validated separately, not the pair), the thermal side requires two units (split to jointly cover the room, rather than one) to match mmWave's room-scale coverage, adding real cost rather than a marginal add-on, and for 2D LiDAR + Thermal specifically, fusing two independent position-sensing streams from scratch was flagged as a disproportionate implementation risk for this project's timeline. Starting with mmWave alone keeps a working single-sensor system as the baseline throughout the phased build, and only asks for the added cost and complexity of a second sensor if the single-sensor result actually falls short.

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